

Bachelor Thesis Exposé

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Title: Verified Quotient Reasoning for
Distance-Rationalizable Voting Rules on the Simplex

1 Introduction

On the one hand, distance-rationalizable (DR) voting rules determine the outcome of an election by finding the closest unambiguous election — where closeness is defined by a *distance measure* and unambiguity by a *consensus class*. Hadjibeyli and Wilson [2019] study the geometric structure of homogeneous and anonymous DR rules by representing election quotients as points on the rational simplex, yielding results on consistency — characterized as convexity of fibers on the simplex — and decisiveness, characterized as the tie-set containing no relatively open subset of the simplex. On the other hand, Diekhoff et al. [2020] develop the Verified Modular Construction Framework (VMCF), which enables the compositional construction of voting rules with machine-verified properties in Isabelle/HOL. Integrating the first with the second strand of research, Steinriede [2022] translates distance rationalization into the VMCF, and Appelhagen [2024] extends this by laying the foundations for working with election quotients: establishing the quotient rule infrastructure, verifying the bijection between homogeneous and anonymous election quotients and the rational simplex, and verifying the lifting of DR distances to their quotients.

The proposed thesis continues this project. Concretely, it pursues three main tasks: instantiating existing DR rules (Kemeny, Borda, Plurality) as verified quotient DR rules within the VMCF (M1); lifting the VMCF’s symmetry composition rules to the quotient level (M2); and connecting quotient DR rules to selected geometric characterisations on the simplex – verifying the characterization of consistency as fiber convexity, as well as decisiveness results, within the VMCF (M3, M4). Two questions guide this work: whether the VMCF’s modular proof infrastructure can be extended to election quotients (Q1), and whether the simplex representation can be integrated into the VMCF in a way that makes selected geometric results available for verified voting-rule constructions (Q2).

2 Background

2.1 Social Choice Theory

Social choice theory studies rules that aggregate individual preferences over alternatives into collective outcomes. In this thesis, the relevant rules are voting rules whose inputs are profiles of linear orders and whose outputs are possibly tied winners or rankings. The central examples are Kemeny, Borda, and plurality, because they are standard rules with known distance-rationalizable presentations.

2.2 Verified construction of voting rules

Much of traditional Social Choice Theory has been about rationalizing the use of certain voting rules by proving that they fulfill desirable properties – axioms like fairness, anonymity, neutrality etc. When voting procedures become more complicated, these become harder to prove. To make this easier, Diekhoff et al. [2020] develop the Verified Modular Construction Framework (VMCF), which allows the composition of different *electoral modules* with verified properties. When building new compositional voting rules within the VMCF, the verified properties of the modules can be preserved, or new ones established, via pre-defined *composition rules*. The idea is to provide building blocks, which are formally verified with the theorem prover Isabelle/HOL to construct complex voting rules that keep the desired properties of their components.

Diekhoff et al. [2020] set up electoral modules with a generalized structure that to defer alternatives to other voting modules. Composition structures include sequential and parallel construction, looping modules with stop conditions, or modules that revise decisions. In this framework, Diekhoff et al. [2020] provide formal verifications in Isabelle/HOL of properties like Condorcet consistency and monotonicity.

2.3 Distance rationalization

The traditional approach of rationalizing – i.e. justifying – a voting rule is axiomatic. Instead of proving that it fulfills certain desiderata, the distance-rationalization of a voting rule R shows that its election result minimizes the individual tradeoffs all voters have to make – i.e. finds the closest consensus class $\mathcal{K}$ defined by a distance measure d .

A voting rule is distance rationalizable (DR) if it can be defined via an appropriate distance measure and consensus class. A lot of voting rules can be redefined in this manner (cf. Meskanen and Nurmi [2008]). Quite intuitively for instance, the Kemeny rule minimizes a distance – defined as the sum of disagreements of the individual preference ranking with the aggregate ranking – to a consensus class – defined as every voter having the exact same ranking.

2.4 Connection of DR voting rules to the simplex

Many voting rules used in practice are both anonymous and homogeneous. For such rules, Hadjibeyli and Wilson [2019] observe that the relevant information about an election can be compressed substantially: since the result depends only on the relative frequency distribution of votes rather than on the identities or ordering of individual voters, each election can be represented as a point on the rational standard simplex $\Delta_M^Q \subset \mathbb{R}^{m!}$, where each coordinate records the proportion of voters submitting a given preference ranking. This yields a bijection between the election quotient $\mathcal{E}/\sim_{\text{hom}}$ and Δ_M^Q , allowing anonymous and homogeneous rules — which Hadjibeyli and Wilson call *simplex rules* — to be studied as functions defined directly on this geometric object.

Hadjibeyli and Wilson develop this connection specifically for the class of *vote-wise* distances, where the distance between two elections is computed by summing up the distances between the individual votes of each voter, defined as ℓ^1 -norm. They show that the quotient of a votewise distance induced in this way corresponds on the simplex to a *Wasserstein* (also called Kantorovich or Earth Mover) distance — a concept well-developed in probability theory, which measures how much “work” is required to transform one distribution into another. This geometric perspective opens two lines of investigation. First, it provides a characterization of *consistency*: a voting rule is consistent — meaning that two sub-electorates agreeing on a winner will continue to agree after combination — if and only if the fibers of its quotient rule on the simplex are convex. Second, it yields results on *decisiveness*, i.e. how often a rule produces a unique winner. Hadjibeyli and Wilson show that rules defined using a 1-votewise distance and a generalized unanimity consensus are *hyperplane rules*, whose tie-sets contain no relatively open subset of the simplex and are therefore negligibly small.

2.5 Integration of DR rules and VMCF

Integrating DR rules into the VMCF stands to reason because consensus classes and distances with known desirable properties can often yield voting rules with desirable properties (cf. Brandt et al. [2016]), providing a new way of the modular composition of voting rules. Steinriede [2022] takes the first step in this endeavor by formalizing the DR framework in Isabelle/HOL. Appelhagen [2024] builds directly on this foundation, turning to the geometric perspective of Hadjibeyli and Wilson. She formalizes symmetry properties of voting rules via group actions and extends the VMCF’s inference rules to cover further symmetry properties of DR rules. Crucially, she establishes the infrastructure for working with election quotients within the VMCF and defining some properties that allow the analogy to the simplex.

3 Research Gap and Motivation

This thesis aims at continuing the project of integrating the results about DR rules obtained from their simplex representation into the VMCF. Appelhagen [2024] lays a base for this undertaking, achieving the following in regard to Hadjibeyli and Wilson [2019]:

- Definition of a general quotient rule infrastructure and integration of election quotient rule into the VMCF.
- Verified proof of the bijection between homogenous and anonymous quotient rules and the standard rational simplex.
- Verified proof of Hadjibeyli and Wilson [2019]’s Lemma 3.15: Under appropriate invariance conditions, the distance measure of a DR rule can be lifted cleanly to its quotient.
- Proof of Hadjibeyli and Wilson [2019] Remark 7.2 (not yet verified): Consistency of a voting rule is convexity of the fibers of its quotient rule on the simplex.
- Definition of a notion of decisiveness on DR rules.

What remains open is the integration of the existing quotient-simplex correspondence as a reusable proof infrastructure. In particular, existing DR rules have not yet been systematically instantiated as quotient DR rules inside the VMCF, and the framework’s composition and inference rules have not yet been lifted to the quotient level. Moreover, central simplex-level characterisations from Hadjibeyli and Wilson [2019], such as consistency as convexity of fibres and decisiveness via the structure of tie sets, have not yet been connected to verified VMCF constructions.

The thesis therefore focuses on two questions:

1. Q1: Can the VMCF’s modular proof infrastructure be extended to work on election quotients?
2. Q2: Can selected simplex-level characterisations of anonymous and homogeneous DR rules be formalised and connected to quotient rules in the VMCF?

To keep the project feasible within the scope of a Bachelor thesis, I propose the following contribution targets (which I will expand on in the next section): The minimal successful outcome is to extend Appelhagen [2024]’s quotient infrastructure from foundational constructions to reusable verified quotient-level reasoning for specific DR voting rules. Concretely, to formalize quotient instantiations for at least two DR rules and prove at least one lifted quotient-level inference rule. The target outcome additionally includes formalising consistency

as convexity of fibres and the infrastructure for decisiveness. Results on Wasserstein distances and the full results on decisiveness are treated as extension goals, depending on the amount of additional formal infrastructure required.

4 Project Plan

4.1 Tasks

(M1) Quotient DR Instances for Kemeny, Borda, and Plurality

Details: For each of the Kemeny rule, the Borda rule, and the plurality rule (to be defined first) following Hadjibeyli and Wilson [2019] Example 2.17, instantiate Appelhagen [2024]’s `distance_R_Q` by: (i) showing `votewise.distance_swap 1_one` is simple w.r.t. `anon_homR` (ii) showing the respective consensus is closed under `anon_homR`; (iii) showing the rule is invariant under `anon_homR`, connecting to Appelhagen [2024]’s homogeneity lemmas (A.7.5); (iv) applying `invar_dr_simple_dist_imp_quotient_dr` to obtain the quotient equality theorem. Delivered artefacts are new theories `Quotient_Kemeny_Rule.thy`, `Quotient_Borda_Rule.thy`, and `Plurality_Rule.thy`, each containing the respective verified quotient instantiation theorem. Whether Copeland falls within or outside the quotient framework is investigated.

Quality Objective: All quotient instantiation theorems proved without sorry.

Main Task Questions: (Q1)

Repository: `Quotient.Distance.Rationalization.thy`;
`Distance.Rationalization.thy`; `Votewise.Distance.Rationalization.thy`;
`Voting_Rules/Kemeny_Rule.thy`; `Voting_Rules/Borda_Rule.thy`;
`Voting_Rules/Plurality_Rule.thy` (new);
`Voting_Rules/Copeland_Rule.thy`

(M2) Lift DR Composition Rules to Quotient Rules

Details: For anonymity, neutrality and reversal symmetry – separately or via a unified group-action approach – state and prove the lifted inference rule: if `distance_R(K, d)` has property P , d is simple w.r.t. `anon_homR`, and $\mathcal{K}$ is closed under `anon_homR`, then `distance_R_Q(anon_homR, d, K)` also has property P . The proof connects Appelhagen [2024] A.11.3 with the symmetry lemmas in `Distance.Rationalization.Symmetry.thy`. Delivered artefacts are lifted inference rules `[ANON'_Q]`, `[NEUTR_Q]`, `[REV_Q]` in a new theory `Quotient.Distance.Rationalization.Properties.thy`, together with corollary instantiations for the Kemeny and Borda quotient

rules from (M1).

Quality Objective: At least two lifted inference rules and one concrete corollary proved without `sorry`.

Main Task Questions: (Q1)

Repository: `Distance.Rationalization.Symmetry.thy`;
`Quotient.Distance.Rationalization.thy`; `Voting.Symmetry.thy`;
`Quotient.Distance.Rationalization.Properties.thy` (new)

(M3) **Consistency as Fibre Convexity**

Details: Add the predicate `consistent` to `Property.Interpretations.thy`: a rule R is consistent if for all disjoint-voter elections E_1, E_2 with $R(E_1) \cap R(E_2) \neq \emptyset$, we have $R(E_1 \cup E_2) = R(E_1) \cap R(E_2)$. Define `convex.fibers` for quotient rules on `vote.simplex`: for all x, y in the fibre of outcome r and all rational $\lambda \in [0, 1]$, the point $\lambda x + (1 - \lambda)y$ lies in the same fibre. Rigorously state, sketch, and formally both directions (HW2019 Remark 7.2, Appelhagen [2024] Example 11) with Isabelle/HOL: convex fibres imply consistency, by formalising the calculation that the combined electorate maps to a rational convex combination of the two sub-electorate simplex points, and the other way around. Delivered artefacts are `consistent` in `Property.Interpretations.thy` (or new `Consistency.thy`), `convex.fibers` and the theorem `convex.fibers_imp_consistent` in `Quotient.Distance.Rationalization.thy`.

Quality Objective: `convex.fibers_imp_consistent` proved without `sorry`.

Main Task Questions: (Q2)

Repository: `Property.Interpretations.thy`;
`Quotient.Distance.Rationalization.thy`; `Election.Quotients.thy` (pre-requisite)

(M4) **Decisiveness as Hyperplane Containment of the Quotient Rule Boundary**

Details: Define `decisiveness_Q` on `distance_R_Q`: the tie set contains no relatively open subset of `vote.simplex`. Prove the equivalence with Appelhagen [2024]’s profile-level `decisiveness` via `invar_dr_simple_dist_imp_quotient_dr`. Define the Votewise Minimizer Property (vmp, HW2019 Definition 6.1) and generalised unanimity consensus (HW2019 Definition 6.4); prove `strong_unanimity` and

weak_unanimity are instances. Prove HW2019 Proposition 6.3 for $p = 1$: under VMP and a 1-votewise distance, the bisector between consensus fibres $\mathcal{K}_r$ and $\mathcal{K}_{r'}$ satisfies $\sum_t x_t \delta(t, r) = \sum_t x_t \delta(t, r')$, a linear equation in the simplex coordinates (From my assessment, this should be achievable without topology.). Optionally, define **hyperplane_rule** and state HW2019 Corollary 6.9: if $\mathcal{K}$ is a generalised unanimity consensus and d is 1-votewise, then **distance_R_Q anon_homR d K** is a hyperplane rule. In this case it is investigated whether the topological conclusion (a hyperplane has empty relative interior in the simplex) can be cited from external Isabelle libraries, or has to be left for future work. Delivered artefacts are **decisiveness_Q**, **vmp**, **generalised_unanimity_consensus**, and **prop_6_3** in **Distance_Rationalization.Symmetry.thy** or a new theory, together with **hyperplane_rule** and Corollary 6.9 in **Quotient.Distance_Rationalization.thy**.

Quality Objective: **decisiveness_Q**, the equivalence theorem, and **prop_6_3** proved without **sorry**; **hyperplane_rule** and, optionally, Corollary 6.9 stated.

Main Task Questions: (Q2)

Repository: **Distance_Rationalization.Symmetry.thy**; **Quotient.Distance_Rationalization.thy**; **Property_Interpretations.thy** (reference); Isabelle Analysis library (Step 5 only)

(A1) ℓ^1 -Votewise Distance as Wasserstein Distance (HW2019 Proposition 3.21)

Details: Apply Appelhagen [2024]’s A.11.1 infrastructure to show **votewise_distance swap 1_one** satisfies the simplification of HW2019 Proposition 3.15. Define the normalised 1-votewise distance d_*^1 (scaled by $1/n$) and verify its homogeneity (HW2019 Remark 3.20). Define the 1-Wasserstein distance d_W^1 on **vote_simplex** via a coupling formulation (HW2019 Definitions 3.16–3.17) in a new theory **Wasserstein.Distance.thy**. Prove $d_*^1 = d_W^1$ on **vote_simplex** (HW2019 Proposition 3.21 for $p = 1$ based on this; the restriction to $p = 1$ keeps all reasoning in rational arithmetic). Delivered artefacts are **Wasserstein.Distance.thy** and the theorem **swap_votewise_quotient_eq_wasserstein**.

Quality Objective: If **implemented** – **swap_votewise_quotient_eq_wasserstein** proved without **sorry** for $p = 1$.

Main Task Questions: (Q2)

Repository: **Quotient.Distance_Rationalization.thy**;

Votewise_Distance_Rationalization.thy; Wasserstein_Distance.thy (new); Election_Quotients.thy (prerequisite)

(A2) Formal Exploration of Hom/Anon Rules Outside ℓ^p
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Details: Survey anonymous and homogeneous voting rules that are DR but whose distance is not ℓ^p-votewise. Dodgson's rule, which is not homogenous, but can be "homogenized" (cf. HW2019 Section 7), might be an interesting starting point. If any such rule is found with verified or verifiable properties, explore what distance it would induce on the simplex. Findings are documented in the thesis discussion chapter regardless of whether an Isabelle theory is produced.

Quality Objective: If implemented – At least one interesting edge case rule identified and analysed in writing. Optionally, formal expressions of the relation of the class of ℓ^p-votewise rules to the class of hom/anon rules.

Additional Task Questions: (Q2)

Repository: –

4.2 Thesis Schedule and Risk Assessment

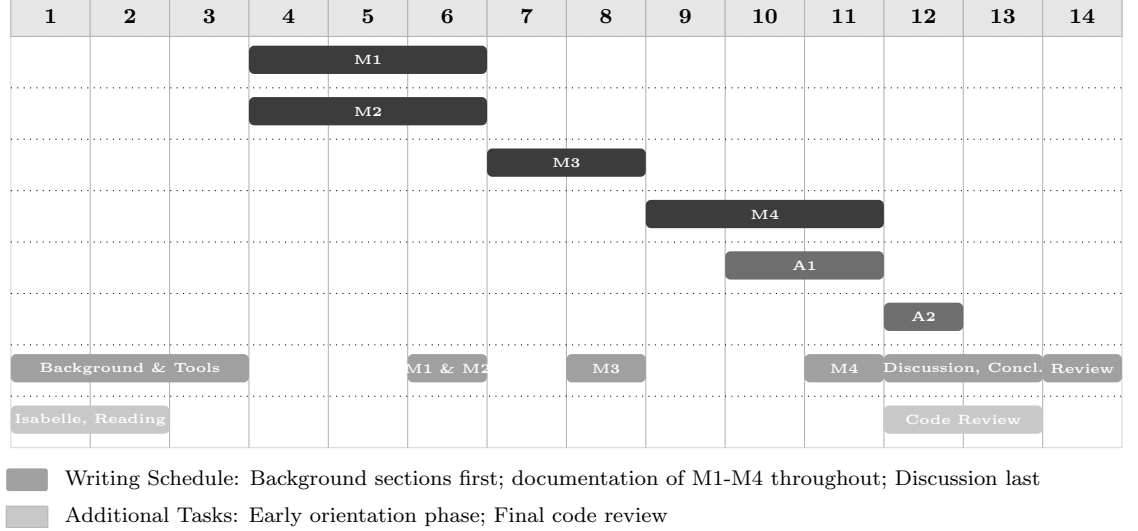


Figure 1: 14-week thesis work schedule.

The schedule is structured along four phases: An orientation period (W1-W3), a period concerning the modules that extend election quotients (M1, M2; W4-W6), a period concerning the modules concerning verifying properties on the simplex (M3, M4, A1; W7-W11) and a finalizing period (A2; W12-W14) concerned with optional modules, writing, and reviews.

The goal is to adhere to these broad phases as strictly as possible, with the possibility of adjustments within the phases concerning two contingencies:

- **Q1 Phase – Dependency of M1 and M2:** I am not yet clear on the relation of the two, and whether they can be possibly combined. A simultaneous implementation will allow for flexibility to integrate new insights.
- **Q2 Phase – Scope:** I am unsure about the doability of my planned time scope for M3, M4 and A1, partially because of my lack of experience with Isabelle/HOL, partially because the uncertainties that come with the rigorous formalization of sometimes vaguely stated formal propositions of Hadjibeyli and Wilson [2019]. A flexible approach concerning weeks 7 – 11 is thus planned. I decided to flag A1 as optional, and limit my ambitions regarding the integration of external libraries in M4.

5 Bibliography

References

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